



Research article

Epigenetic mechanism of LncRNA KCNQ1OT1 in high-glucose-induced podocyte injury via m6A methylation modification

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ABSTRACT

Podocyte injury is a hallmark of diabetic nephropathy associated with proteinuria and renal dysfunction. This study elucidated the role of long non-coding RNA KCNQ1 overlapping transcript 1 (LncRNA KCNQ1OT1) in high-glucose (HG)-induced podocyte injury. In HG-treated podocytes, KCNQ1OT1, WTAP, and MAPK6 were quantified. After KCNQ1OT1 interference, cell viability, apoptosis, inflammation, and oxidative stress were measured. Subcellular localization of KCNQ1OT1 was determined by nuclear-cytoplasmic fractionation and RNA FISH. Binding of MLL1 to KCNQ1OT1 was validated via RNA pull-down and RIP. MLL1/H3K4me3 enrichment on the WTAP promoter was detected by ChIP. m6A levels of MAPK6 were measured by MeRIP. Binding of IGF2BP1 to MAPK6 was confirmed via RIP. Rescue experiments explored the WTAP/MAPK6 axis in podocyte injury. HG treatment upregulated KCNQ1OT1, WTAP, and MAPK6 in podocytes. KCNQ1OT1 knockdown enhanced cell proliferation, reduced apoptosis, and attenuated inflammation and oxidative stress. Mechanistically, nuclear-localized KCNQ1OT1 recruited MLL1 to the WTAP promoter, enhancing H3K4me3 modification and WTAP activation. WTAP promoted m6A methylation of MAPK6 mRNA, which was stabilized by IGF2BP1. Over-expression of WTAP or MAPK6 abrogated the protective effects of KCNQ1OT1 knockdown on podocytes injury. In conclusions, KCNQ1OT1 exacerbates HG-induced podocyte injury by upregulating the WTAP/MAPK6 axis in an m6A-dependent manner.

1. Introduction

Diabetic nephropathy (DN), a prominent and severe complication of diabetes mellitus (DM), has experienced a steep global rise in both incidence and prevalence [1]. The vicious cycle of oxidative stress-inflammation triggers pathological damage to renal cells in DN pathology, including podocyte injury [2]. Podocytes are terminally differentiated epithelial cells whose slit diaphragms determine the glomerular filtration rate and selective filtration of macromolecules [3]. Damaged podocytes lead to compromised integrity of the glomerular filtration barrier, thereby increasing proteinuria and abnormal glomerular filtration rate [4]. Clinically, DN management includes pharmacological interventions for glycemic control, blood pressure reduction, and lipid regulation, while podocyte-targeted therapies have shown promising strategy [5]. In this study, we established a DN cell model by treating podocytes with high glucose (HG), aiming to identify new

mechanisms underlying podocyte injury in DN.

Long non-coding RNAs (LncRNAs) function as important epigenetic regulators that modulate chromatin organization, RNA transcription, and splicing, and are involved in mRNA transport, stability, and translation, as well as protein stability and post-translational modifications [6]. Abnormal upregulation of LncRNAs is considered an important manifestation of DN pathology, including renal fibrosis, oxidative stress, and mitochondrial damage [7]. LncRNA Kcnq1 overlapping transcript 1 (KCNQ1OT1) shows high expression in HG-induced glomerular mesangial cells and endothelial cells [8]. Of note, silencing KCNQ1OT1 can alleviate HG-induced podocyte apoptosis and suppress inflammation [9]. LncRNA recruits the histone methyltransferase mixed lineage leukemia 1 (MLL1) to the promoter regions of genes, induces tri-methylation of lysine 4 on histone H3 (H3K4me3), and activates gene transcription [10]. However, existing studies have focused on the mechanism of KCNQ1OT1 in DN through the ceRNA mechanism [11],

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and no previous studies have explored the involvement of KCNQ1OT1 in podocyte injury via histone modification.

N6-methyladenosine (m6A) is an internal cotranscriptional modification added by m6A methyltransferases, removed by demethylases, and recognized by m6A binding proteins, which is ubiquitously present in eukaryotic RNAs [12]. Wilms tumor 1-associated protein (WTAP) serves as a critical constituent of the m6A methyltransferase complex that participates in m6A modification by recruiting the m6A methyltransferase complex to target mRNAs [13]. During DN progression, WTAP is highly expressed and promotes apoptosis, mitochondrial damage, and inflammatory responses in an m6A-dependent manner [14–16]. Notably, the WTAP promoter harbors H3K4me3 modification and is regulated by histone modifications [17]. We hypothesize that LncRNA KCNQ1OT1 recruits MLL1 to the WTAP promoter and influences H3K4me3 modification, thereby participating in the pathology of DN.

The mitogen-activated protein kinase (MAPK) cascade represents a highly conserved signaling pathway across species, playing a pivotal role in transmitting extracellular signals into various cellular responses to maintain homeostasis [18]. MAPK6 is one of the atypical members of the MAPK family, which exhibits increased expression in DN and contributes to podocyte injury [19–21]. Database prediction indicates that MAPK6 contains m6A modification sites, and thus we hypothesize that MAPK6 is a downstream target of WTAP.

In this study, we aim to explain the mechanism of the LncRNA KCNQ1OT1/WTAP/MAPK6 axis in the development of DN through podocyte injury, and provide more theoretical insights for DN treatment.

2. Materials and methods

2.1. Cell culture and treatment

The mouse podocyte cell line MPC5 from the American Type Culture Collection (ATCC, Manassas, VA, USA) were maintained in RPMI 1640 medium supplemented with 10 % bovine serum, 100 U/mL penicillin/streptomycin, and 10 U/mL recombinant mouse IFN- γ (Sigma, St. Louis, MO, USA) at 37 °C in a 5 % CO₂ incubator. In glucose treatment, 5 mM glucose (NG group) or 30 mM glucose (HG group) were treated with cells for 24, 48, or 72 h. The NG group was additionally treated with 25 mM mannitol (Sigma) to establish the control osmolarity [15].

2.2. Cell transfection

siRNAs targeting KCNQ1OT1, MLL1, or IGF2BP1 (si-KCNQ1OT1, si-MLL1, si-IGF2BP1) and pcDNA3.1 plasmids for WTAP or MAPK6 overexpression were designed and synthesized by GenePharma (Suzhou, Jiangsu, China) and Hanbio (Shanghai, China). Cells were seeded in 6-well plates until reaching 70 % confluency and transfected by employing Lipofectamine 3000 (Invitrogen, Carlsbad, CA, USA). The final concentration of siRNA was 20 nM. The sequences of siRNAs are shown in Table 1.

Table 1
siRNA sequences.

Name	Sequences
si-KCNQ1OT1-1 (si-KCNQ-1)	SS: GCGUGGUUACGGUCUAGUACA AS: UACUAGACCGUAAACACGCCG
si-KCNQ-2	SS: GGAUCACUUGAGGACUACACC AS: UGUAGUGCUCAAGUGAUCCGA
si-KCNQ-3	SS: CGUGGUCUAUGAUGGUGCAUU AS: UGCACCAUCAUAGACCACGCG
si-MLL1	SS: CCAGGUCUCCUUCAGUUAAG AS: UUAACUGAAGGAGACCGUGUG
si-IGF2BP1	SS: GGAGCUCGCGACACUUCAGG AS: UGAAGUUGUCGGAGCUCGCG

2.3. Cell counting kit-8 (CCK-8) assay

Cell viability was evaluated using a CCK-8 kit (C0042, Beyotime, Shanghai, China). Briefly, cells (5×10^3 cells/well) were seeded in 96-well plates. Subsequently, addition of 10 μ L of CCK-8 solution was conducted in each well, followed by incubation at 37 °C for an additional 2 h. Absorbance was measured at 450 nm on a microplate reader (Thermo Fisher Scientific, Waltham, MA, USA).

2.4. 5-ethynyl-2'-deoxyuridine (EdU) staining

Cell proliferation was determined via an EdU assay. Briefly, cells were incubated with EdU solution (C0071S, 50 μ M, Beyotime) for 5 h, fixed in 4 % paraformaldehyde (Beyotime), permeabilized using 1 % Triton X-100 (Sigma, St. Louis, MO, USA), stained with staining solution, and underwent counterstaining with 4',6-diamidino-2-phenylindole (DAPI), followed by observation under a fluorescence microscope (Olympus, Tokyo, Japan), with EdU-positive cells labeled by green fluorescence and nuclei by blue fluorescence.

2.5. Flow cytometry

Cell apoptosis was evaluated with an Annexin V-FITC/PI apoptosis detection kit (C1062L, Beyotime). Briefly, 1×10^6 cells were cultured for 48 h, washed, and resuspended in binding buffer. Cells were then incubated with 10 μ L of Annexin V-FITC and 5 μ L of PI in the dark for 15 min, and apoptosis rates were analyzed by flow cytometry (cytoFlex S, Beckman, Miami, FL, USA).

2.6. Enzyme-linked immunosorbent assay (ELISA)

Cell supernatants were collected after centrifugation at 2500g for 15 min at 4 °C, and 50 μ L of supernatant was used to measure tumor necrosis factor (TNF)- α (ab285327, Abcam, Cambridge, MA, USA), interleukin (IL)-6 (ab222503, Abcam), and IL-1 β (ab229440, Abcam) levels using ELISA kits. Measurement of absorbance at 450 nm was conducted on a Bio-Rad microplate reader.

2.7. Lactate dehydrogenase (LDH), malondialdehyde (MDA), and superoxide dismutase (SOD) assays

Levels of LDH and MDA and SOD activity were evaluated using corresponding assay kits according to the manufacturer's protocols. Briefly, cells (2×10^5 cells/well) were seeded in 6-well plates, and cell supernatants were collected after centrifugation at 14,000 g for 5 min at 4 °C. LDH levels were detected using an LDH assay kit (C0016, Beyotime). MDA levels and SOD activity were measured using an MDA assay kit (ab118970, Abcam) and a SOD colorimetric activity kit (S0101M, Beyotime) respectively.

2.8. Nuclear-cytoplasmic fractionation

RNA was extracted using an RNA subcellular fractionation kit (78833, Thermo Fisher Scientific), and real-time polymerase chain reaction was performed. Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) was used as the endogenous control for cytoplasmic RNA, and U6 served as the control for nuclear RNA.

2.9. RNA fluorescence in situ hybridization (RNA FISH)

The localization of KCNQ1OT1 in cells was analyzed by performing FISH experiments using a FISH kit (c10910, Ribobio, Guangzhou, China). The specific procedures were as follows: Cell climbing slices were placed at the bottom of 24-well plates, and then cells in the logarithmic growth phase were digested and added to the climbing slices (approximately 6×10^4 cells per well). Before the experiment, the cell

adhesion rate was maintained at 60 %–70 %. First, the cells were rinsed with $1 \times$ phosphate-buffered saline (PBS) solution for 5 min, followed by fixation with 4 % paraformaldehyde at room temperature for 10 min. After fixation, the cells were rinsed with $1 \times$ PBS solution for 5 min, and this step was repeated three times. To prepare for hybridization, 1 mL of pre-cooled permeabilization solution was added to each well and incubated at 4 °C for 5 min. Subsequently, the permeabilization solution was removed, and the cells were rinsed with $1 \times$ PBS solution for 5 min, with this step repeated three times. Next, 200 μ L of pre-hybridization solution was added to each well and sealed at 37 °C for 30 min. In a dark environment, 2.5 μ L of FISH Probe mixed storage solution was mixed with hybridization solution. The pre-hybridization solution was removed from each cell well, and then the hybridization solution containing Cy3-labeled KCNQ1OT1 probe was added. Hybridization was performed at 37 °C overnight. To reduce background signal, the cells in the wells were washed three times with Wash Solution I for 5 min each time, followed by one wash with Wash Solution II and one wash with Wash Solution III respectively. Subsequently, the cells were washed with $1 \times$ PBS at room temperature for 5 min under dark conditions, and then stained with DAPI staining solution for 10 min. The cell climbing slices were carefully separated from the wells, fixed on coverslips with mounting medium, and the samples were scanned and photographed under an Olympus microscope. The specific probe for KCNQ1OT1 (Ensembl ID: ENSMUSG00000101609) was custom-synthesized by Ribobio.

2.10. RNA pull-down

Binding probabilities between KCNQ1OT1 and MLL1 were predicted using the RPISeq database (<http://pridb.gdcb.iastate.edu/RPISeq/>) [22]. Biotinylated KCNQ1OT1 probes or negative control (NC) probes were designed and synthesized by Ribobio (Guangzhou, Guangdong, China). Briefly, 1×10^7 cells were lysed, and cell lysates (containing RNase inhibitor) were incubated with KCNQ1OT1 probes or NC probes overnight at 4 °C. Complexes were isolated using M – 280 streptavidin-conjugated magnetic beads (s3762, Sigma) pre-coated with BSA and yeast tRNA to prevent RNase activity, separated by polyacrylamide gel electrophoresis (PAGE), and analyzed for MLL1 expression by western blotting.

2.11. RNA immunoprecipitation (RIP)

The EZMagna RIP kit (17–701, Millipore, Billerica, MA, USA) was used to determine interactions between KCNQ1OT1 and MLL1. Cells were lysed in radioimmunoprecipitation assay (RIPA) buffer supplemented with protease inhibitor cocktail and RNase inhibitors. Cell extracts were incubated with magnetic beads conjugated to MLL1 antibody (NB600-248, Novus Biologicals, Littleton, CO, USA), IGF2BP1 antibody (ab313422, Abcam), or immunoglobulin G (IgG) control antibody (ab172730, Abcam) in RIP buffer at 4 °C for 6 h. Beads were washed three times with wash buffer, treated with proteinase K at 55 °C for 30 min, and KCNQ1OT1 enrichment was detected utilizing real-time quantitative polymerase chain reaction (RT-qPCR).

2.12. Chromatin immunoprecipitation (ChIP) assay

ChIP assays were carried out using the EZ-Magna ChIP kit (17–408, EMD Millipore). Cells were treated with 4 % paraformaldehyde and subsequently incubated with glycine for 10 min to induce DNA-protein crosslinks. Cells were lysed with cell lysis buffer and ultrasonicated to generate chromatin fragments. Lysates were co-precipitated with Magnetic Protein A Beads conjugated to MLL1 antibody (NB600-248, Novus Biologicals), H3K4me3 antibody (ab213224, Abcam), or IgG control antibody (ab172730, Abcam). Precipitated DNA was analyzed by RT-qPCR. Enrichment values were calculated relative to the input. Primers are listed in Table 2.

Table 2
PCR primer sequences.

Name	Sequences (5'-3')
LncRNA KCNQ1OT1	F: CGCGATCCTCCTCAGTGT R: CATATCGCGACCATGA
MLL1	F: TTCTAAGGAGGCGGTGGTG R: ACGCTTGTCTGTCTGGATGG
WTAP	F: CTCAGCAAGTTCAGCAGCC R: TGGGCTTGTCTCCAGTTGTCT
IGF2BP1	F: GCTGTAGGTCTCTCCAGC R: ACTTGTACCATCTCTGCTCC
MAPK6	F: GGCAGTACCAGATCACCAC R: TTAACCTCGCTCTCCCTCCA
WTAP promoter	F: CCGGGCTCATAAACCGACT R: CGGGTGAGGGTCTGGAGTA
GAPDH	F: GGTCCCAGCTTAGTTTCATCA R: AATCCGTTACACCGACCTT

2.13. m6A quantification

m6A modification sites in MAPK6 were predicted using the SRAMP database (<http://www.cuilab.cn/m6asiteapp/old>) [23]. m6A RNA methylation was quantified using a colorimetric m6A RNA methylation assay kit (ab185912, Abcam). Briefly, to each well, 80 μ L of binding buffer and 200 ng of sample RNA were introduced, followed by a 90-min incubation at 37 °C to facilitate RNA binding. After three washes with wash buffer, 50 μ L of diluted capture antibody was added to each well, which was then incubated at room temperature for 60 min. Detection antibody and enhancer solution were subsequently added, and plates were incubated at room temperature for 30 min. Finally, development solution was added and incubated in the dark at 25 °C for 1–10 min, followed by reaction termination with stop solution. Absorbance was measured at 450 nm.

2.14. Methylated RNA immunoprecipitation (MeRIP)

mRNA was purified from total RNA using the dynabeads mRNA purification kit (61006, Invitrogen). Following RNA quality analysis by NanoDrop 2000, changes in m6A levels in mRNA were measured using the Magna MeRIP™ m6A Kit (17–10499, Millipore) according to the manufacturer's protocol. A portion of mRNA was reserved as Input, and the remaining mRNA was used for m6A immunoprecipitation. After immunoprecipitation with Magna ChIP protein A/G magnetic beads, samples were eluted twice with elution buffer. Washed lysates were transferred to new tubes and incubated with IgG antibody (ab171870, Abcam) or m6A antibody (ab151230, Abcam) by rotation at 4 °C for 2 h. Immunoprecipitated m6A RNA was recovered by ethanol precipitation, and MAPK6 levels were detected by RT-qPCR.

2.15. Actinomycin D treatment

Cells were treated with 0.2 mM actinomycin D (GlpBio, Montclair, CA, USA) for a 30-min period (designated as 0 h sample). Samples were collected at 2, 4, and 6 h, and total RNA was extracted. MAPK6 mRNA levels were measured by RT-qPCR.

2.16. RT-qPCR

Cells were treated with Trizol (Invitrogen) for RNA extraction, after which cDNA was produced using the QuantiTect Reverse Transcription Kit (205311, Qiagen, Valencia, CA, USA). RT-qPCR was performed on a LightCycler 96 (Roche, Basel, Switzerland). All RNA levels were normalized to GAPDH, and relative expression was calculated using the $2^{-\Delta\Delta Ct}$ method [24]. Primers are listed in Table 1.

2.17. Western blot analysis

After cell collection and lysis in protease inhibitor-supplemented RIPA buffer, protein concentration measurements were performed employing a BCA detection system (23227, Thermo Fisher Scientific). Proteins were separated by 4–10 % sodium dodecyl sulfate-PAGE and transferred to polyvinylidene fluoride membranes, which were blocked with 5 % non-fat milk in tris-buffered saline and tween 20 for 1 h. After blocking, membranes were subjected to incubation with primary antibody at 4 °C, then exposure to horseradish peroxidase-conjugated secondary antibodies (1:2000, ab205718, Abcam). Protein bands were analyzed using an ECL Western blotting substrate (32109, Thermo Fisher Scientific). Primary antibodies included: MLL1 (1:1000, NB600-248, Novus Biologicals), WTAP (1:1000, ab195380, Abcam), IGF2BP1 (1:1000, ab313422, Abcam), MAPK6 (1:1000, ab53277, Abcam), and GAPDH (1:2500, ab9485, Abcam). Original band images are shown in the Supplementary files.

2.18. Statistical analysis

Data were analyzed using SPSS 21.0 (IBM SPSS Statistics, Chicago, IL, USA) and GraphPad Prism 8.0 (GraphPad Software Inc., San Diego, CA, USA) for statistical analysis and plotting. Normality and homogeneity of variance were first tested. For two-group comparisons, Student's t-test was used; for multi-group comparisons, one-way or two-way

analysis of variance (ANOVA) was applied, followed by Tukey's multiple comparisons test. Statistical significance was set at $p < 0.05$, with $p < 0.01$ indicating highly significant differences.

3. Results

3.1. HG induces podocyte injury and promotes LncRNA KCNQ1OT1 expression

As a first step to evaluate the function of LncRNA KCNQ1OT1 in podocytes exposed to HG, KCNQ1OT1 expression was measured at specified time points (0, 24, 48, and 72 h) after HG treatment. As shown in Fig. 1A, KCNQ1OT1 levels in podocytes were significantly upregulated in a time-dependent manner following HG stimulation ($p < 0.01$), peaking at 48 h, indicating a critical role for KCNQ1OT1 in HG-induced podocyte injury. Prolonged HG exposure led to decreased cell viability, reduced proliferation, increased apoptosis rates, and elevated LDH levels ($p < 0.01$, Fig. 1B–E). HG induction increased TNF- α , IL-6, and IL-1 β levels, enhanced MDA production, and decreased SOD activity ($p < 0.01$, Fig. 1F–H). Collectively, HG induces podocyte injury and promotes LncRNA KCNQ1OT1 expression.

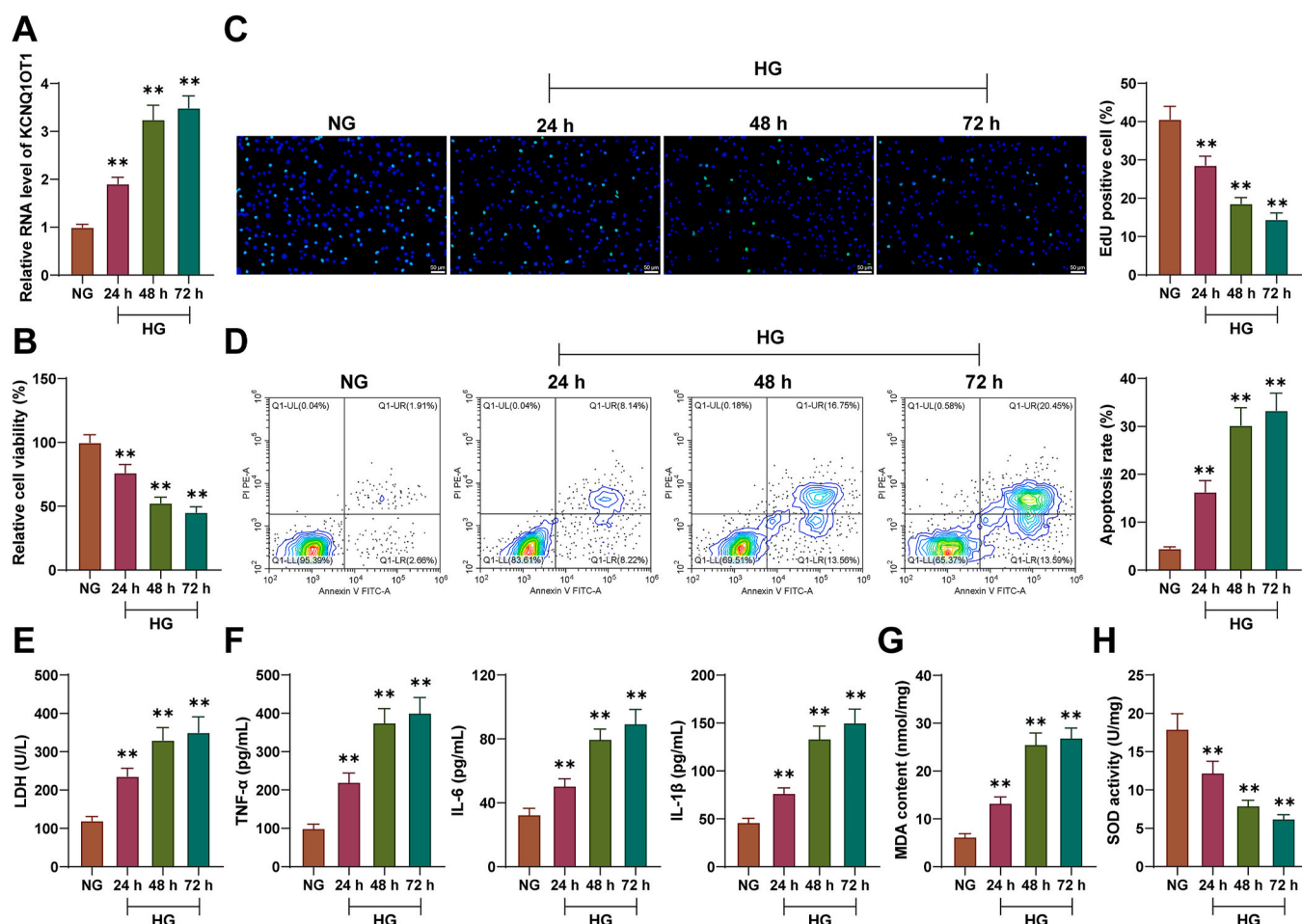


Fig. 1. HG induces podocyte injury and promotes LncRNA KCNQ1OT1 expression. A: The expression of LncRNA KCNQ1OT1 in cells was detected by RT-qPCR; B: Cell viability was measured by CCK-8 assay; C: Cell proliferation was assessed by EdU assay; D: Cell apoptosis was analyzed by flow cytometry; E: LDH levels in cells were detected using an LDH kit; F: Levels of TNF- α , IL-6, and IL-1 β in cells were measured by ELISA; G: MDA levels in cells were detected using an MDA kit; H: SOD activity in cells was measured using an SOD kit. Three biological replicates were performed, and data are presented as mean $\pm$ standard deviation. One-way ANOVA was used for comparisons among multiple groups, followed by Tukey's multiple comparisons test for post hoc analysis. ** $p < 0.01$.

3.2. Knockdown of LncRNA KCNQ1OT1 alleviates HG-induced podocyte injury

Next, siRNA was transfected into podocytes to downregulate KCNQ1OT1 expression ($p < 0.01$, Fig. 2A), and two siRNAs with optimal transfection efficiency were selected for subsequent assays. Following KCNQ1OT1 knockdown, HG-induced reduction in cell viability was reversed, with increased proliferation, decreased apoptosis rates, and lower LDH levels ($p < 0.05$, Fig. 2B–E). Inflammatory and oxidative stress analyses revealed that KCNQ1OT1 knockdown suppressed HG-induced elevation of TNF- α , IL-6, and IL-1 β levels, reduced MDA production, and enhanced SOD activity ($p < 0.05$, Fig. 2F–H). Collectively, KCNQ1OT1 knockdown alleviates HG-induced podocyte injury.

3.3. LncRNA KCNQ1OT1 recruits MLL1 to WTAP and enhances H3K4me3 modification to activate WTAP

KCNQ1OT1 was predominantly localized in the nucleus (Fig. 3A–B). LncRNAs can recruit MLL1 to gene promoters to induce H3K4me3 modification and activate gene transcription [10]. The RPISeq database predicted a high binding probability between KCNQ1OT1 and MLL1 (Fig. 3C). WTAP is upregulated in diabetic nephropathy [15] and undergoes H3K4me3 modification [17], but whether LncRNA KCNQ1OT1 recruits MLL1 to the WTAP promoter to affect H3K4me3 modification remains unclear. RNA pull-down and RIP assays confirmed the interaction between KCNQ1OT1 and MLL1 ($p < 0.01$, Fig. 3D–E).

Subsequently, ChIP results showed enrichment of MLL1 and H3K4me3 on the WTAP promoter, which was reduced following KCNQ1OT1 knockdown ($p < 0.05$, Fig. 3F). Knockdown of MLL1 alone also decreased MLL1 and H3K4me3 enrichment on the WTAP promoter and reduced WTAP expression ($p < 0.05$, Fig. 3F–H). WTAP expression was upregulated in podocytes after HG stimulation but decreased following KCNQ1OT1 knockdown ($p < 0.01$, Fig. 3I–J). In conclusion, LncRNA KCNQ1OT1 applies MLL1 into the WTAP promoter, induces H3K4me3 modification, and activates WTAP expression.

3.4. Overexpression of WTAP attenuates the protective effect of LncRNA KCNQ1OT1 knockdown on HG-induced podocyte injury

Subsequently, pcDNA3.1-WTAP was transfected into podocytes to upregulate WTAP expression, followed by co-transfection with si-KCNQ1OT1 (si-KCNQ-1) for combination experiments ($p < 0.01$, Fig. 4A–B), prior to HG induction. WTAP overexpression decreased cell viability, increased LDH levels, reduced proliferation, and elevated apoptosis rates ($p < 0.05$, Fig. 4C–F). Compared with KCNQ1OT1 knockdown alone, the combination group exhibited increased levels of TNF- α , IL-6, IL-1 β , and MDA, as well as decreased SOD activity ($p < 0.05$, Fig. 4G–I). These results indicate that WTAP overexpression alleviates the protective effect of LncRNA KCNQ1OT1 knockdown against HG-induced podocyte injury.

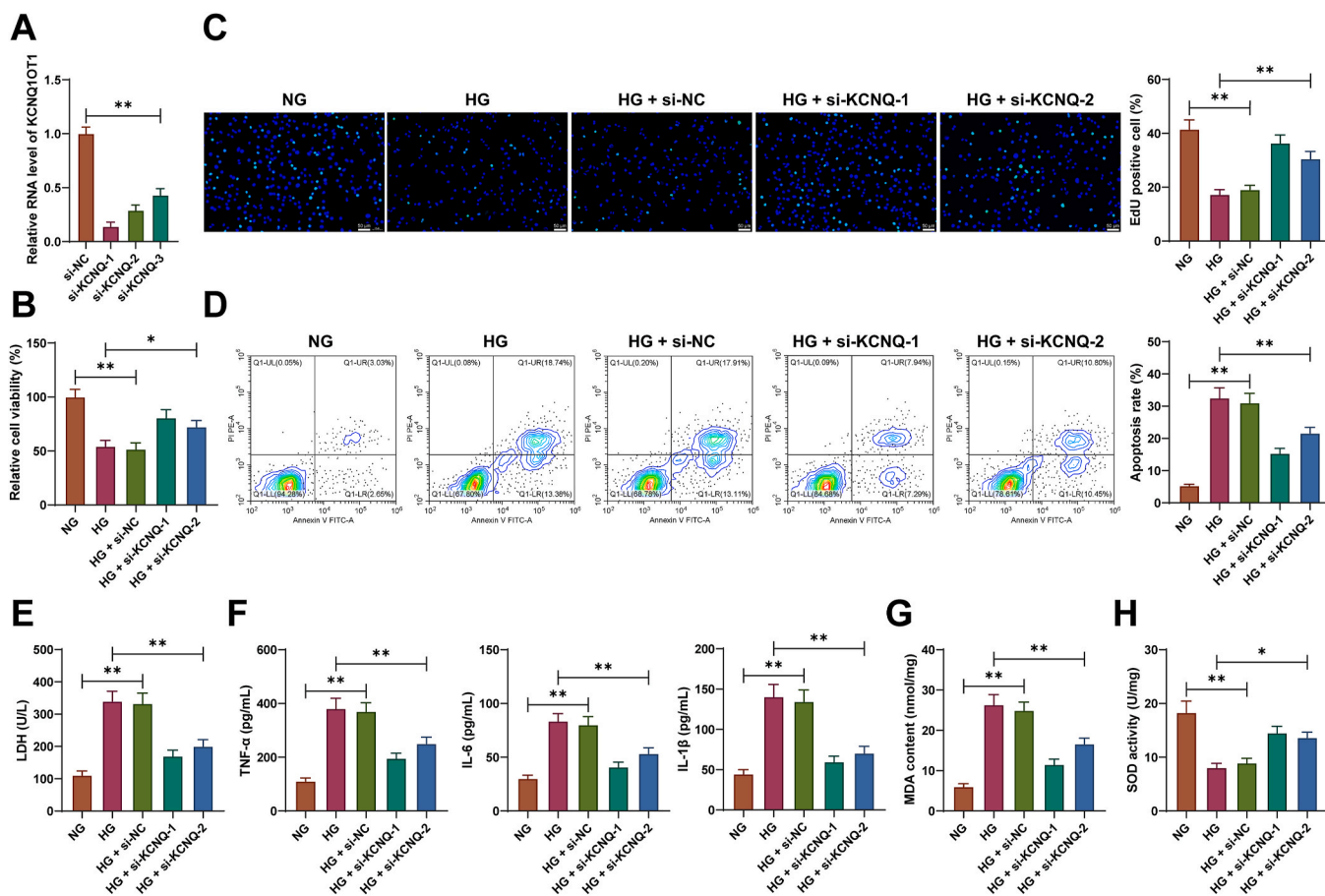


Fig. 2. Knockdown of LncRNA KCNQ1OT1 alleviates HG-induced podocyte injury. Specific siRNA targeting KCNQ1OT1 (si-KCNQ1OT1) was transfected into podocytes by researchers, with NC siRNA (si-NC) transfection serving as the negative control. A: The transfection efficiency of LncRNA KCNQ1OT1 was detected by RT-qPCR; B: Cell viability was measured by CCK-8 assay; C: Cell proliferation was assessed by EdU assay; D: Cell apoptosis was analyzed by flow cytometry; E: Intracellular LDH levels were detected by an LDH kit; F: TNF- α , IL-6, and IL-1 β levels in cells were measured by ELISA; G: Intracellular MDA levels were detected by an MDA kit; H: Intracellular SOD activity was measured by an SOD kit. Three biological replicates were conducted, and data are presented as mean $\pm$ standard deviation. Comparisons among multiple groups were performed by one-way ANOVA, followed by Tukey's multiple comparisons test. * $p < 0.05$, ** $p < 0.01$.

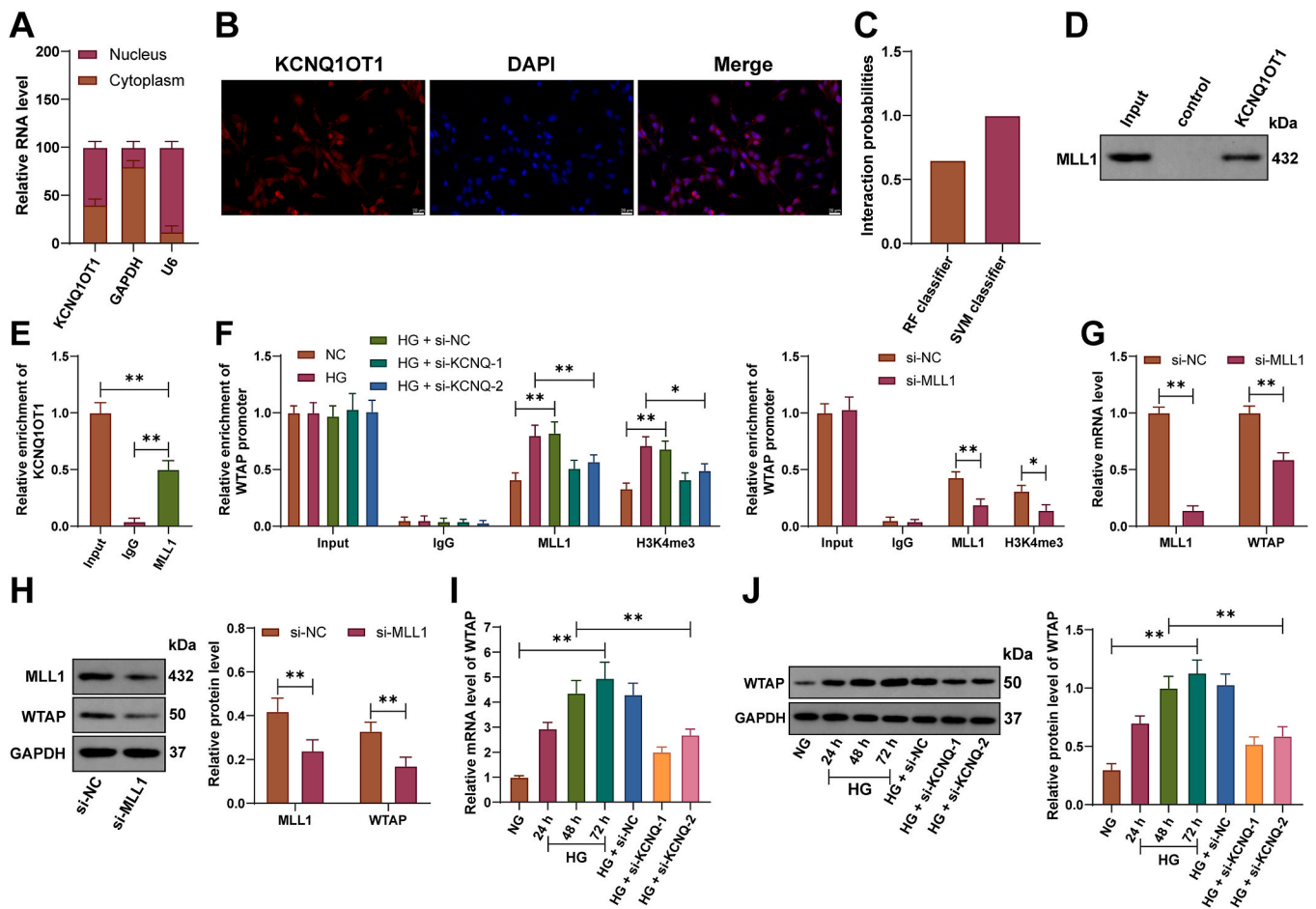


Fig. 3. LncRNA KCNQ1OT1 recruits MLL1 to the WTAP promoter and induces H3K4me3 modification to activate WTAP. A–B: Subcellular localization of KCNQ1OT1 was detected by nuclear-cytoplasmic fractionation and RNA FISH; C: The binding probability between KCNQ1OT1 and MLL1 was predicted by the RPISeq database; D–E: The binding of KCNQ1OT1 to MLL1 was detected by RNA pull-down and RIP assays; F: Enrichment of MLL1 and H3K4me3 on the WTAP promoter was detected by ChIP assay; G–H: Expressions of MLL1 and WTAP in cells were detected by RT-qPCR and Western blot; I–J: WTAP expression in cells was detected by RT-qPCR and Western blot. Three biological replicates were performed, and data are presented as mean $\pm$ standard deviation. For data in panels E and I–J, comparisons among multiple groups were analyzed by one-way ANOVA; for data in panels F–H, comparisons among multiple groups were analyzed by two-way ANOVA, followed by Tukey's multiple comparisons test. * $p < 0.05$, and ** $p < 0.01$.

3.5. WTAP enhances m6A methylation of MAPK6 mRNA and upregulates MAPK6 stability via IGF2BP1

WTAP regulates ATF4 mRNA expression by influencing its m6A methylation [25]. Given that MAPK6 is upregulated in diabetic nephropathy [19–21] and the SRAMP database predicts m6A modification sites in MAPK6 (Fig. 5A), we hypothesized that MAPK6 is a downstream target of WTAP. Global m6A levels were increased after HG stimulation, decreased upon KCNQ1OT1 knockdown, and were restored after WTAP overexpression ($p < 0.01$, Fig. 5B). Subsequently, MeRIP assays showed m6A modification on MAPK6 mRNA, which was reduced after KCNQ1OT1 knockdown and enhanced after WTAP overexpression ($p < 0.01$, Fig. 5C). WTAP mediates IGF2BP1-stabilized m6A modification of NLRP3 mRNA [15]. Our further experiments revealed that IGF2BP1 was enriched on MAPK6 mRNA, with enrichment trends consistent with m6A modification levels ($p < 0.01$, Fig. 5D). WTAP overexpression alone increased MAPK6 mRNA stability and protein expression ($p < 0.01$, Fig. 5G–I), whereas IGF2BP1 knockdown ($p < 0.01$, Fig. 5E–F) attenuated MAPK6 mRNA stability and expression ($p < 0.05$, Fig. 5G–I). MAPK6 expression was upregulated by HG stimulation, decreased upon KCNQ1OT1 knockdown, and rescued after WTAP overexpression ($p < 0.01$, Fig. 5H–I). WTAP promotes m6A methylation of MAPK6 mRNA and enhances MAPK6 stability via IGF2BP1.

3.6. Overexpression of MAPK6 attenuates the protective effect of LncRNA KCNQ1OT1 knockdown on HG-induced podocyte injury

Finally, pcDNA3.1-MAPK6 was transfected into podocytes to upregulate MAPK6 expression, followed by co-transfection with si-KCNQ1OT1 (si-KCNQ-1) for combination experiments ($p < 0.01$, Fig. 6A–B), followed by HG induction. MAPK6 overexpression decreased cell viability, increased LDH levels, reduced proliferation, and elevated apoptosis rates ($p < 0.01$, Fig. 6C–F). Compared with KCNQ1OT1 knockdown alone, MAPK6 overexpression caused significant increases in TNF- α , IL-6, IL-1 β , and MDA levels, as well as decreased SOD activity ($p < 0.01$, Fig. 6G–I), thereby attenuating the protective effect of LncRNA KCNQ1OT1 knockdown against HG-induced podocyte injury.

4. Discussion

DN is a highly prevalent complication of diabetes that can be modulated by epigenetic mechanisms through histone modifications and non-coding RNAs [26]. Extensive studies have confirmed that LncRNAs exhibit potential as therapeutic targets by participating in major pathological processes of diabetes, such as oxidative stress, inflammation, and fibrosis [27]. Here, we discovered an epigenetic mechanism in which LncRNA KCNQ1OT1 activates WTAP transcription through H3K4me3 histone modification, and WTAP promotes MAPK6

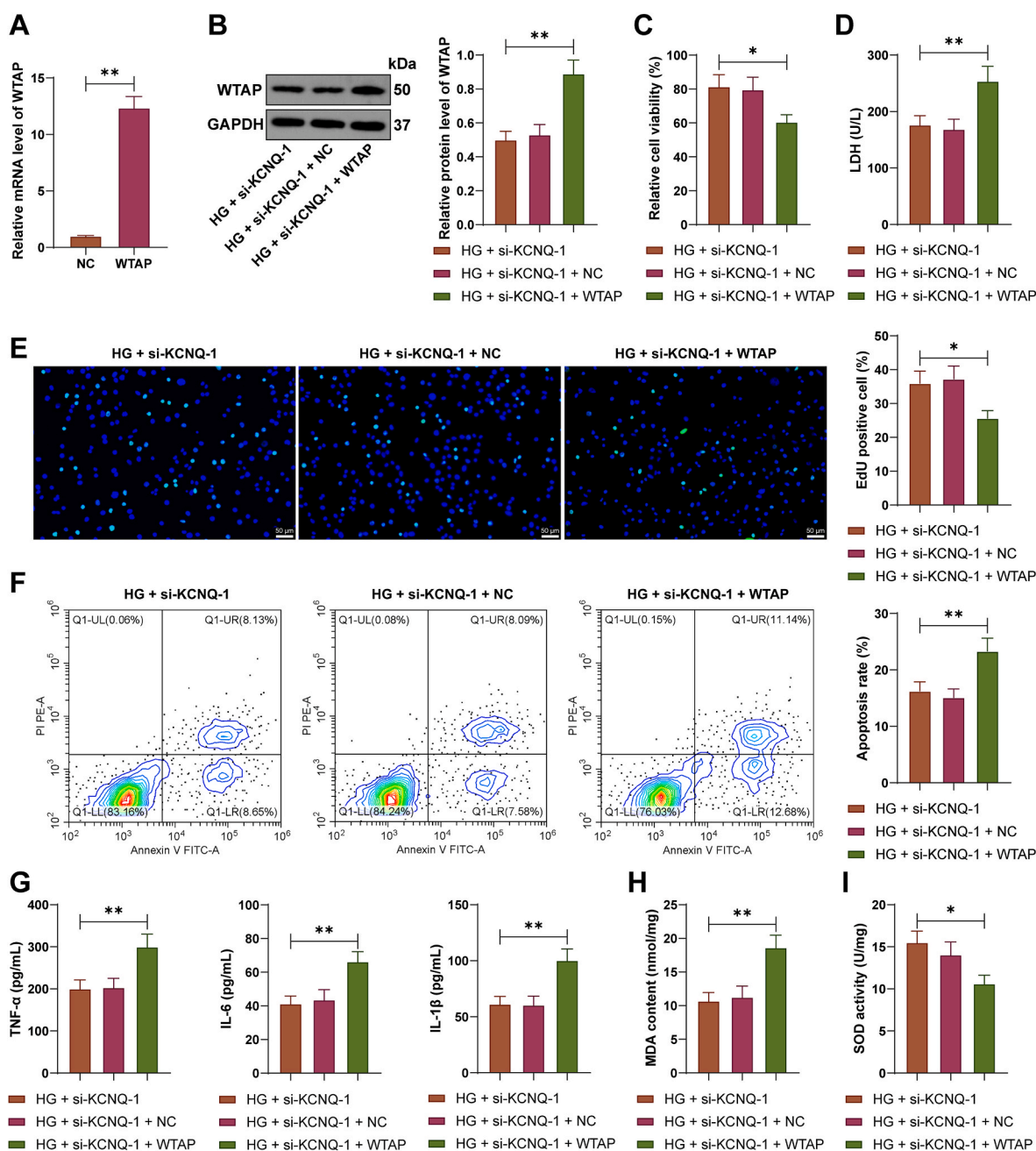


Fig. 4. Overexpression of WTAP attenuates the protective effect of LncRNA KCNQ1OT1 knockdown on HG-induced podocyte injury. WTAP pcDNA3.1 (WTAP) was transfected into podocytes by researchers, with NC pcDNA3.1 (NC) transfection serving as the negative control. A: The transfection efficiency of WTAP in cells was detected by RT-qPCR; B: WTAP expression in cells was detected by Western blot; C: Cell viability was measured by CCK-8 assay; D: Intracellular LDH levels were detected by an LDH kit; E: Cell proliferation was assessed by EdU assay; F: Cell apoptosis was analyzed by flow cytometry; G: Levels of TNF- α , IL-6, and IL-1 β in cells were measured by ELISA; H: Intracellular MDA levels were detected by an MDA kit; I: Intracellular SOD activity was measured by an SOD kit. Three biological replicates were performed, and data are presented as mean $\pm$ standard deviation. Data comparisons between the two groups in panel A were performed by *t*-test; comparisons among multiple groups in panels B–I were analyzed by one-way ANOVA, followed by Tukey's multiple comparisons test. **p* < 0.05, and ***p* < 0.01.

expression in an m6A-dependent manner, ultimately exacerbating HG-induced podocyte injury (Fig. 7).

LncRNAs are key regulators in the pathological process of DN, including renal cell injury and inflammation [28]. For example, knockout of LncRNA EVF-2 in DN mice alleviates proteinuria and reduces podocyte injury and inflammation [29]. Knockout of LncRNA Dlx6os1 inhibits HG-induced fibrosis and inflammatory factor release in glomerular mesangial cells [30]. Knockout of LncRNA MALAT1 in DN renal tissues and HG-stimulated HK-2 cells attenuates apoptosis and cell injury [31]. Here, in HG-induced podocytes, we found that KCNQ1OT1

expression was upregulated. Following knockdown of KCNQ1OT1, HG-induced podocyte injury was alleviated. Consistently, another report has shown that KCNQ1OT1 inhibition alleviates the sponging effect on miR-23b-3p and restricts sema3A expression, thus mitigating inflammation and apoptosis in HG-induced podocyte [9]. Knockdown of KCNQ1OT1 restores miR-18b expression to inhibit HG-induced proliferation of human mesenchymal cells, reduce MDA, increase SOD activity, and decrease extracellular matrix accumulation [32]. KCNQ1OT1 overexpression may lead to glomerular mesangial cell proliferation and thickening of the glomerular capillary basement membrane in DN rats

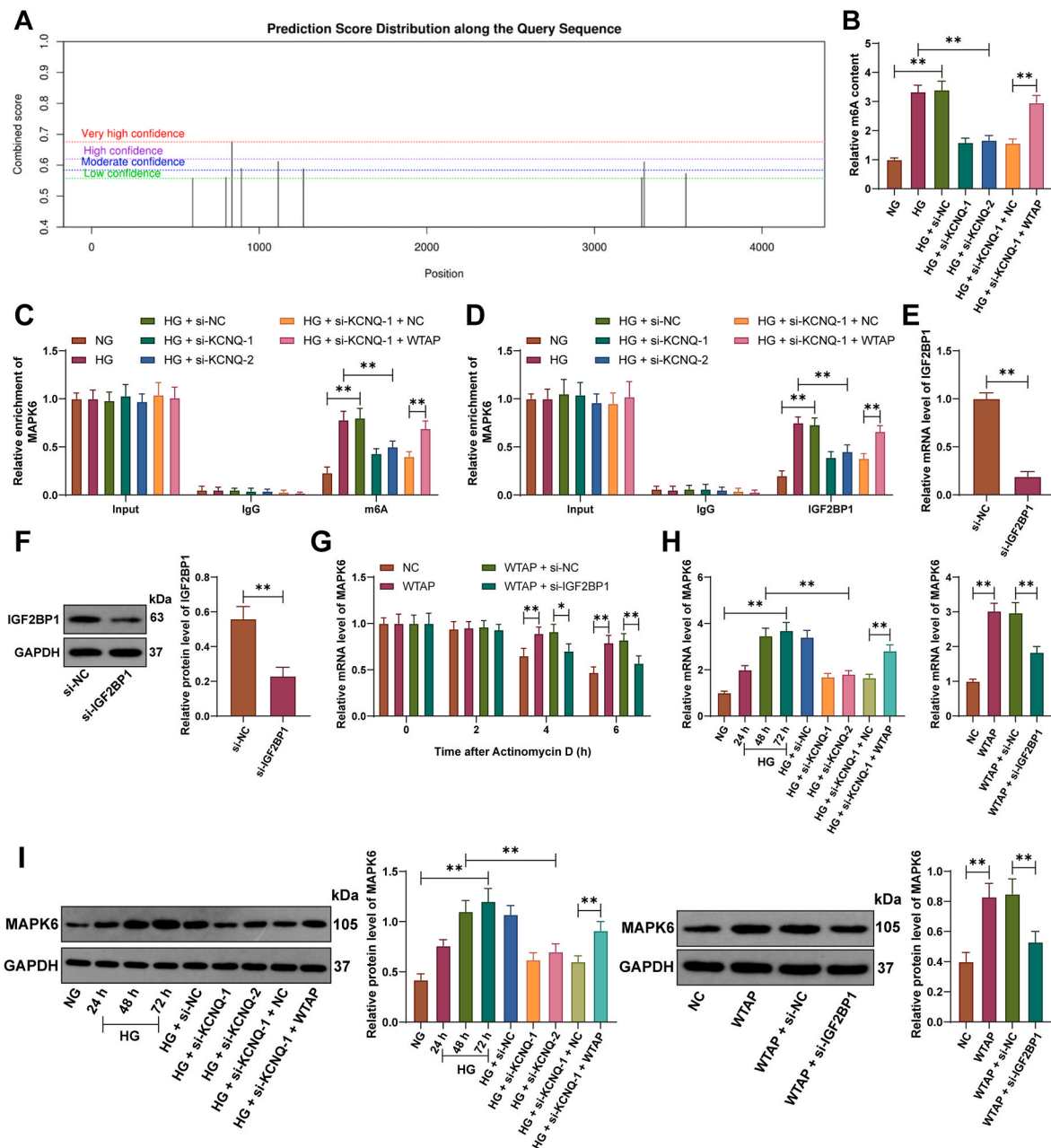


Fig. 5. WTAP enhances m6A methylation of MAPK6 mRNA and upregulates MAPK6 stability via IGF2BP1. A: m6A modification sites on MAPK6 were predicted by the SRAMP database; B: Global m6A levels in cells were quantified by m6A quantification assay; C–D: Enrichment of m6A or IGF2BP1 on MAPK6 was detected by RIP assay; E–F: IGF2BP1 expression in cells was detected by RT-qPCR and Western blot; G: MAPK6 mRNA stability was measured by RT-qPCR; H–I: MAPK6 expression in cells was detected by RT-qPCR and Western blot. Three biological replicates were performed, and data are presented as mean $\pm$ standard deviation. Data comparisons between the two groups in panels E–F were analyzed by *t*-test; comparisons among multiple groups in panels B and H–I were analyzed by one-way ANOVA; comparisons among multiple groups in panels C–D and G were analyzed by two-way ANOVA, followed by Tukey's multiple comparisons test. **p* < 0.05, and ***p* < 0.01.

by activating the MEK/ERK signaling pathway [33]. Additionally, silencing KCNQ1OT1 reduces the expression of pyroptosis-related factors, inhibits caspase-1 activation, and alleviates HG-induced diabetic keratopathy [34]. In diabetic cardiomyopathy, silencing KCNQ1OT1 inhibits the TGF- β 1/Smad pathway, thereby alleviating fibrosis and improving cardiac function [35]. Taken together, KCNQ1OT1 acts as a pathogenic factor for diabetes and its complications.

Following knockdown of KCNQ1OT1, WTAP expression was down-regulated. In the context of HG-induced podocyte injury, we up-regulated WTAP expression and observed increased inflammation. In HG-treated HK-2 cells, WTAP expression is activated via H3K27ac

modification, thereby promoting m6A methylation of NLRP3 mRNA and mediating pyroptosis and pro-inflammatory cytokine release [15]. Overexpression of WTAP mediates m6A modification of ENO1 and promotes its expression, thereby reducing DN cell viability and promoting the secretion of inflammatory factors [14]. Notably, WTAP promoted the m6A modification of downstream target in a manner dependent on the m6A reader IGF2BP1, thereby exacerbating cell apoptosis and mitochondrial dysfunction in DN [16]. Herein, we found that WTAP promoted podocyte injury by enhancing MAPK6 mRNA stability and expression through IGF2BP1-dependent m6A modification. Moreover, substantial evidence has confirmed that IGF2BP1 serves as a

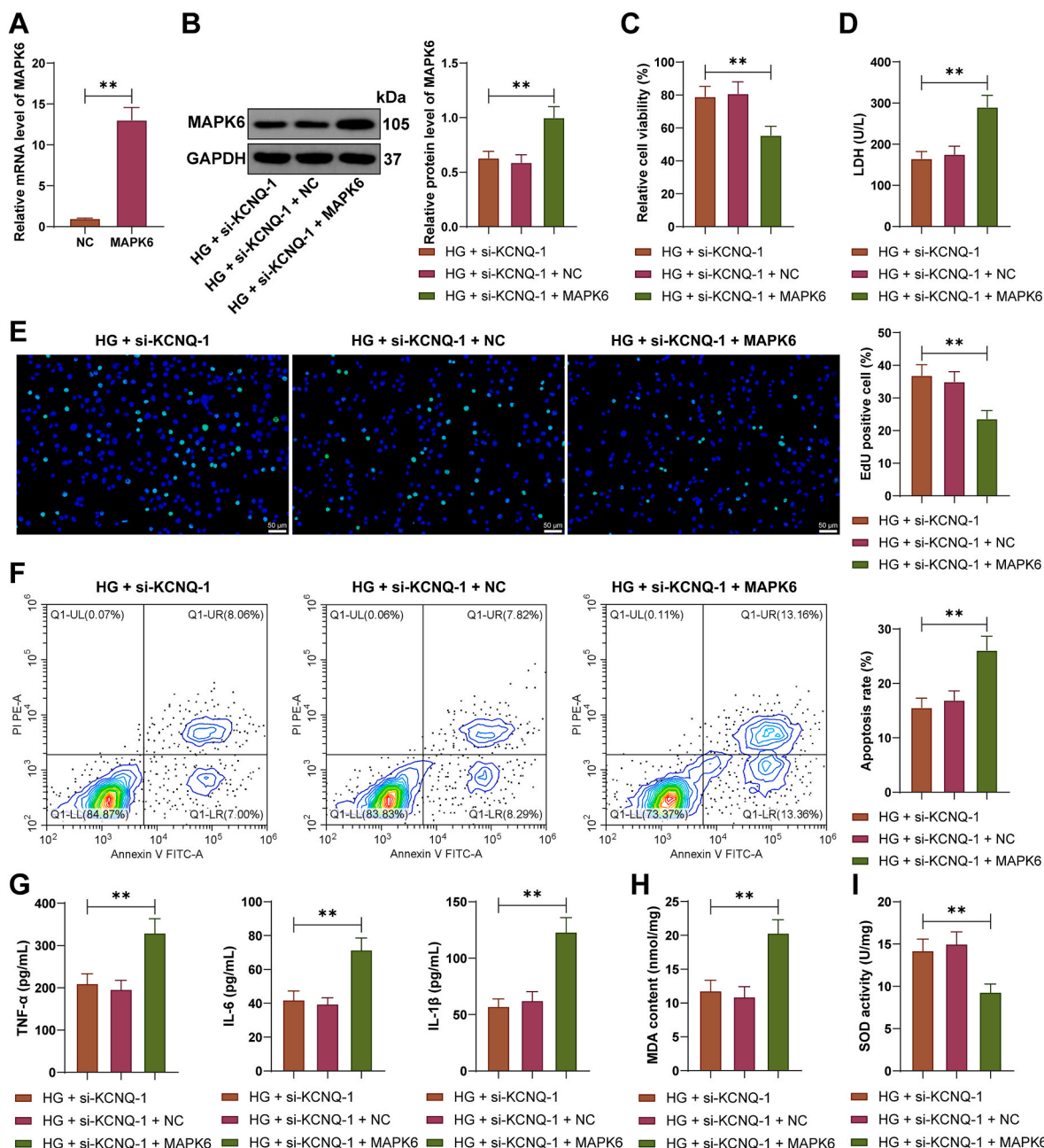


Fig. 6. Overexpression of MAPK6 attenuates the protective effect of LncRNA KCNQ1OT1 knockdown on HG-induced podocyte injury. MAPK6 pcDNA3.1 (MAPK6) was transfected into podocytes, with transfection of NC pcDNA3.1 (NC) serving as the negative control. A: Transfection efficiency of MAPK6 in cells was detected by RT-qPCR; B: MAPK6 expression in cells was detected by Western blot; C: Cell viability was measured by CCK-8 assay; D: LDH levels in cells were detected using an LDH kit; E: Cell proliferation was assessed by EdU assay; F: Cell apoptosis was analyzed by flow cytometry; G: Levels of TNF-α, IL-6, and IL-1β in cells were measured by ELISA; H: MDA levels in cells were detected using an MDA kit; I: SOD activity in cells was measured using an SOD kit. Three biological replicates were conducted, and data are presented as mean ± standard deviation. Data between the two groups in panel A were compared using a *t*-test; data among multiple groups in panels B–I were compared using one-way ANOVA, followed by Tukey's multiple comparisons test. ***p* < 0.01.

key m6A reader in the progression of DN [36,37]. Understanding the molecular mechanisms of WTAP in m6A methylation via IGF2BP1 provides novel research instructions for DN treatment.

In this study, MAPK6 expression in podocytes was significantly upregulated after HG stimulation. A previous study has shown that activation of MAPK6 promotes oxidative stress injury and inflammatory response in podocytes [19]. During the progression of DN, the activation pathway of MAPK6 primarily involves circRNAs acting as sponges for downstream miRNAs to relieve the inhibition of MAPK6. Circ_0000285 promotes podocyte injury by sponging miR-654-3p and activating MAPK6, thereby increasing the levels of IL-6, IL-1β, and TNF-α in DN

cells [21]. After HG treatment, circ_0003928 in HK-2 cells targets and inhibits miR-31-5p, increases MAPK6 expression, and thereby induces cell apoptosis, inflammation, fibrosis, and oxidative stress [20]. MAPK6 is a central regulator of lipolysis in adipocytes that inhibits energy dissipation by forming a complex with MK5 and promotes the output of fatty acids from adipose tissue, thereby facilitating the development of obesity and diabetes [38]. In addition, MAPK6 can regulate hormone-sensitive lipase and monoglyceride lipase, suggesting its important role in metabolic health and diabetes [39]. Regarding the downstream mechanism of MAPK6, previous reports have indicated that enhanced MAPK6 expression activates the MAPK signaling pathway [40,

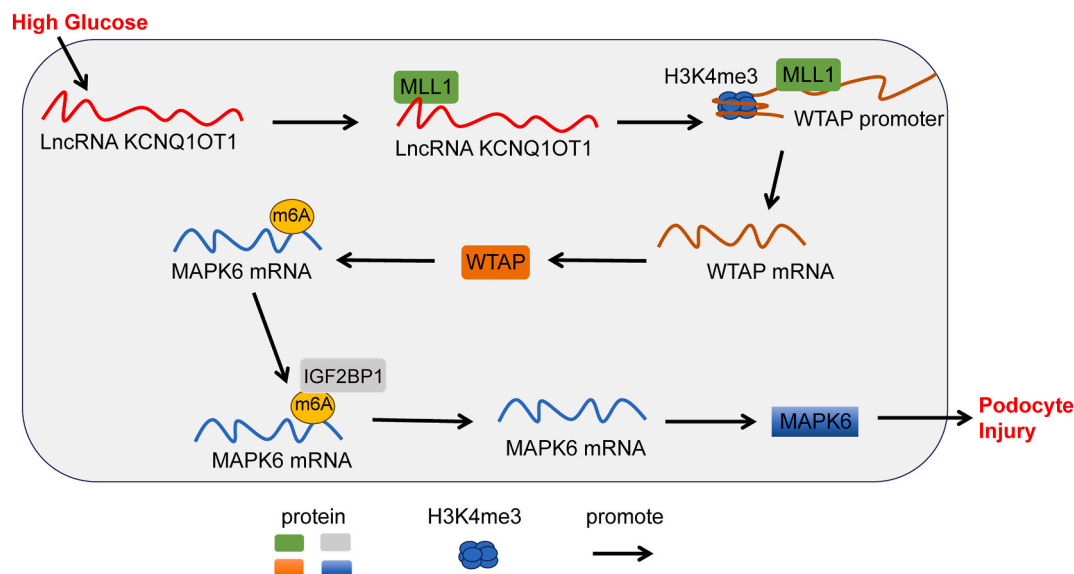


Fig. 7. The epigenetic mechanism by which LncRNA KCNQ1OT1 promotes HG-induced podocyte injury. LncRNA KCNQ1OT1 recruits MLL1 to the WTAP promoter, induces H3K4me3 modification, activates WTAP transcription, and promotes WTAP expression, increases m6A methylation modification of MAPK6 mRNA, upregulates MAPK6 mRNA stability via IGF2BP1 and enhances MAPK6 expression, and ultimately promotes HG-induced podocyte injury.

41]. Furthermore, MAPK/ERK is involved in diabetic pathology by exacerbating oxidative stress and inflammation [42]. Consistently, MAPK6 is associated with Akt phosphorylation and activates Akt, which is a key signaling pathway in insulin resistance-induced obesity and diabetes [43–45]. Therefore, investigating the downstream mechanisms of MAPK6 in DN may provide more theoretical basis.

Although LncRNA KCNQ1OT1 has been studied in DN, our study provides a new mechanism for KCNQ1OT1. Nevertheless, we acknowledge that some limitations still exist in this study. We only performed cell-based experiments, and our proposed mechanism has not been validated in vivo. We have not explored the in-depth mechanisms of WTAP as a component of the m6A methyltransferase complex, such as other m6A reader proteins besides IGF2BP1. In future studies, we plan to deeply probe into the role of WTAP in DN, compare the effects of LncRNA KCNQ1OT1 and other unrelated LncRNAs on HG-induced podocytes, investigate the downstream mechanisms of MAPK6 in DN, and explore the upstream of WTAP to identify new therapeutic targets for DN.

In summary, LncRNA KCNQ1OT1 upregulates WTAP through histone modification, thereby increasing MAPK6 expression and promoting HG-induced podocyte injury.

CRediT authorship contribution statement

Yikai Hou: Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Danli Wang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Lijun Liu:** Visualization, Investigation, Conceptualization. **Jiakun Tian:** Methodology, Investigation, Formal analysis. **Min Zhi:** Supervision, Methodology, Conceptualization. **Yongbing Shi:** Supervision, Methodology, Formal analysis.

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Declaration of competing interest

Authors state no conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found at <https://doi.org/10.1016/j.yexcr.2026.114897>.

Data availability

Data will be made available on request.

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